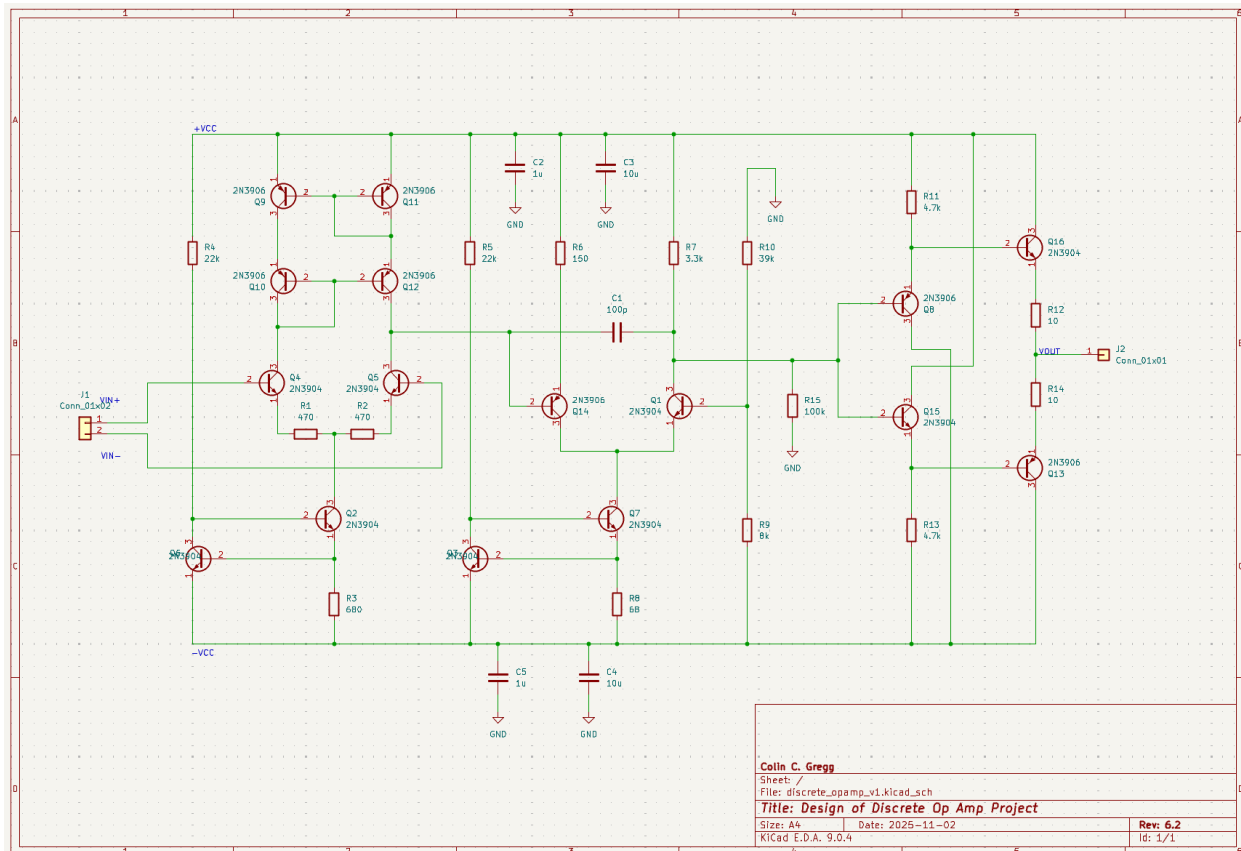


Design of a Personal Discrete Op Amp

[Report is still in progress]

Colin Gregg

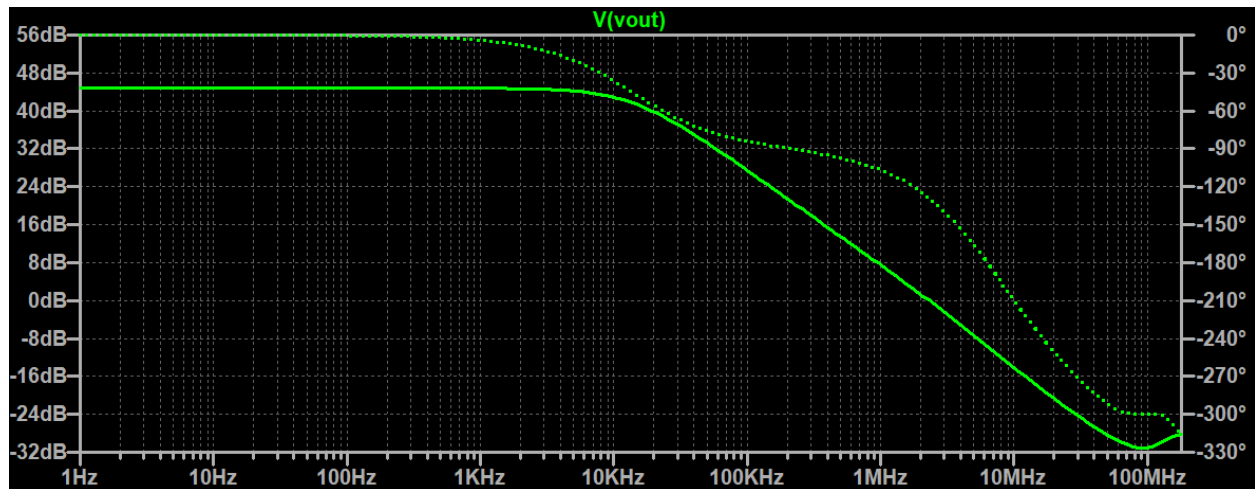
11/2/2025



Results

I am still in the process of writing the full report, however, I want to upload something to show the current progress of the project. I have a simulated working result in LTSpice, and breadboarded it to mixed results: the gain was working correctly, however, the input signal from the signal generator became extremely noisy when connected to the op amp. I am working on a PCB layout to see if this will fix the issues.

Open-loop Transfer Function:



Specifications Compared to LM741:

Spec	My Op Amp	LM741
Gain	0.178 V/mV (45db)	200 V/mV (106dB)
GBP	2.3 MHz	1.5 MHz
R_in (Kilo ohms)	334 kΩ	2 – 6 MΩ
R_out (ohms)	54 Ω	60 Ω
Slew Rate (V/us)	3 V/μs (@ unity gain)	0.5 V/μs (@ unity gain)
CMRR*		90 dB
PSRR*		96 dB
Power Consumption*		50 mW

* still to be determined in simulation

Verification of results:

Gain Bandwidth Product

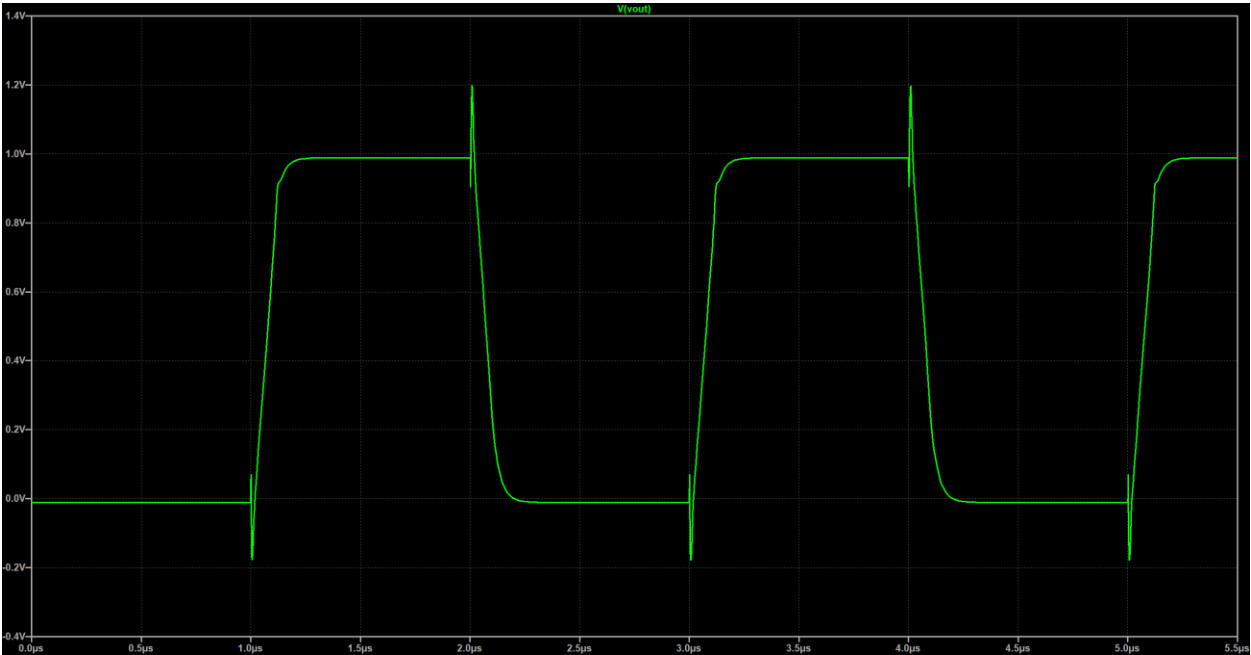
Measured from open-loop transfer function shown above.

Gain (dB)	Gain (times)	Bandwidth (kHz)	GBP
32	39.8107	58	2309.0206
24	15.8489	150	2377.335
16	6.309573	373	2353.4707
8	2.511886	945	2373.7323
1	1.122018	2000	2244.036

Step Response

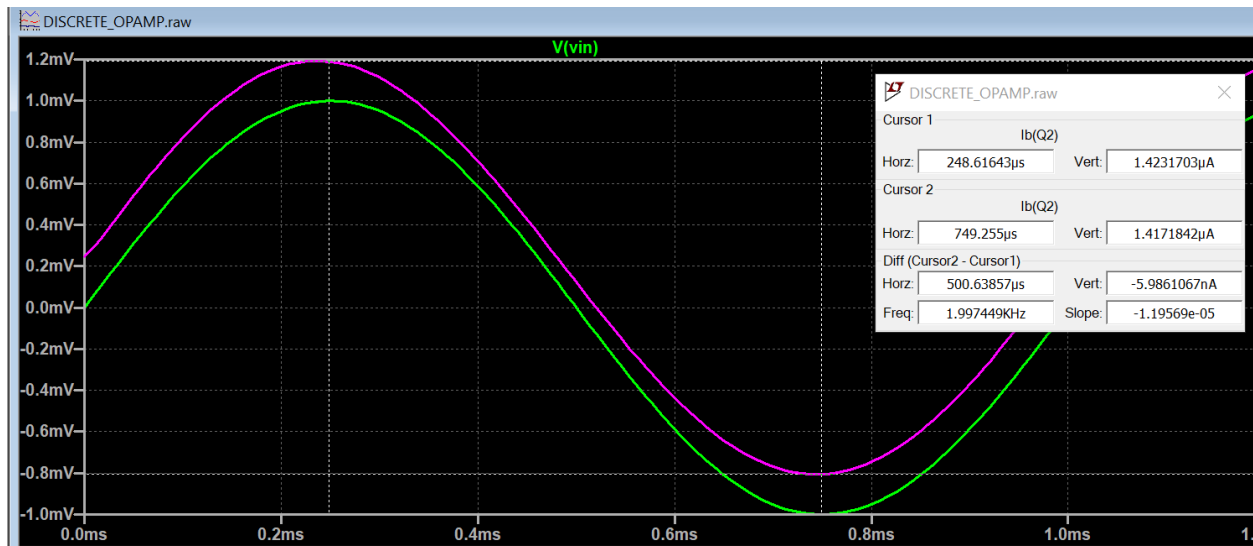
1ns rise and fall times.

20% overshoot



Input Impedance

Measured in open-loop configuration.



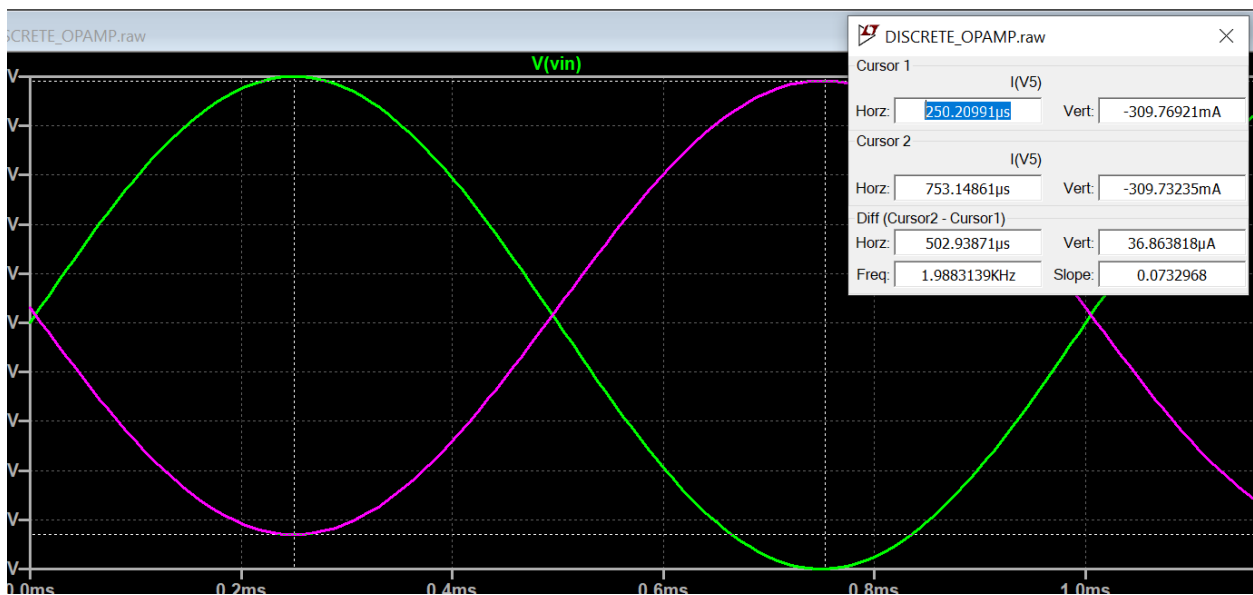
Input = 1 mV $\rightarrow$ dV = 2 mV

dI = 5.9861067 nA

$Z_{in} = dV/dI = 334.11 \text{ k}$

Output Impedance

Measured in open-loop configuration.



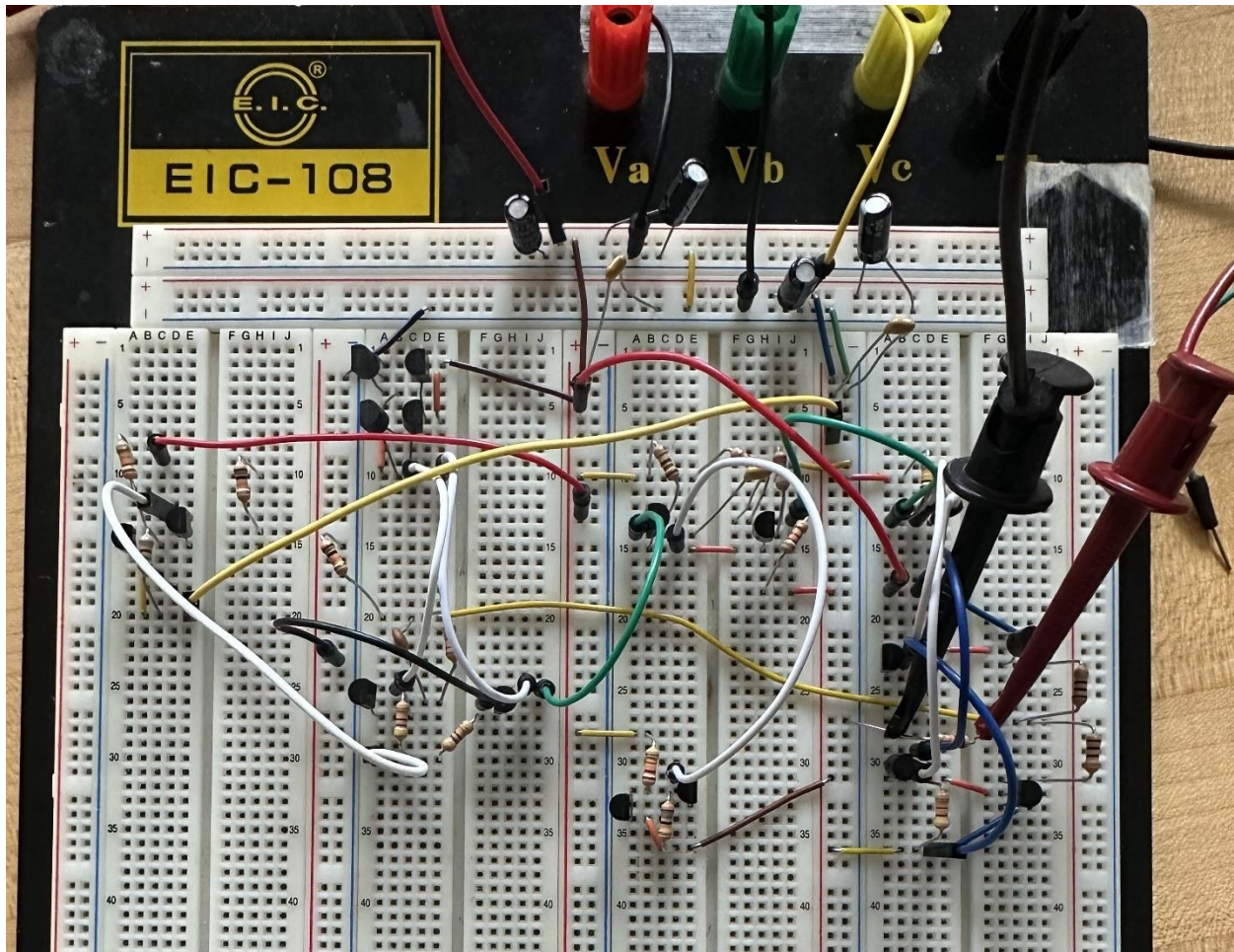
Input = 1 mV $\rightarrow$ dV = 2 mV

dI = 36.863818 μ A

$Z_{out} = dV/dI = 54.25 \text{ ohms}$

Breadboard design

Setup

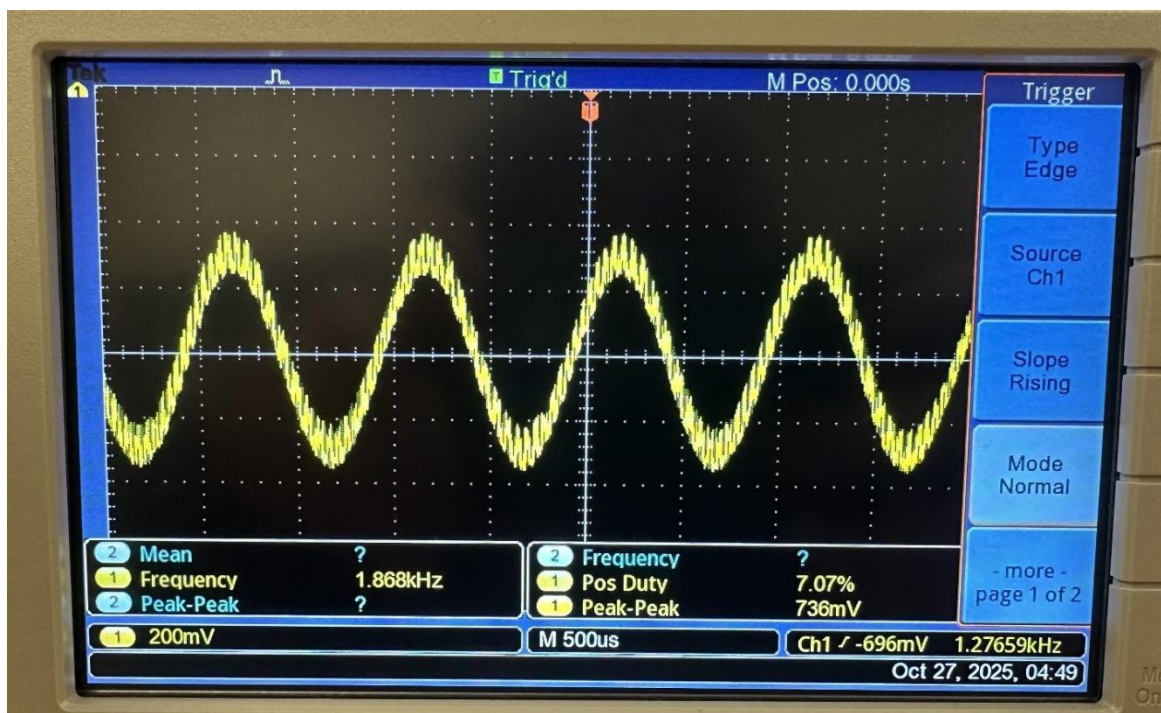


Results

200mV, 25kHz, NFB Ratio = 33k : 10k



200mV, 1kHz, NFB = 33k : 10k



Issues:

- Extremely noisy on both input & output
- Measuring signal generator directly still gives an extremely noisy/fuzzy signal

Next Steps

- PCB (in progress)
 - To reduce noise from using breadboard & get proper results
- Design for more specific use (ex. Very low noise for audio)
 - Not a general-purpose op amp design
 - Use more specific parts too, not only 2N3904/2N3906's
- Improvements that could be made
 - Higher open loop gain
 - Increase Z_{in} of VAS to increase the gain of the diff amp stage (reduce current loss from "loading" of output of diff amp)
 - Higher Input resistance
 - Currently have a large break frequency for open-loop transfer function... maybe get it so it is smaller?
 - Current gain is $>0\text{dB}$ when phase goes below -180°